Stanford CS224v Course

Agentic AI

# Lecture 7

# Data-driven Deep Research

**Shicheng Liu** & Monica Lam

# Lecture Goal

- **Open-ended data-driven research over databases**
  - Build on the agentic approach from SPINACH and DataTalk
  - How to choose research questions, analyze results, and write a report?
- **Automatic reproduction and robustness checks for data-driven deep research**
  - How to automatically conduct reproduction and robustness checks on existing studies?

# Outline

- **DataSTORM: Data-driven deep research**
  - Motivation and running example
  - The agentic approach
  - Evaluation
- SEER-Repro: Automatic robustness checking
  - Reproduction and robustness
  - Results and expert review

# The ACLED Database

Armed Conflict Location & Event Data · one row per recorded event

1.63M rows × 31 columns (2020–2025)

## Schema (selected)

| | |
|---|---|
| **Time / place** | event_date (date); country, admin1, admin2, location (text) |
| **Event** | event_type, sub_event_type (text) |
| **Actors** | actor1, actor2, assoc_actor_1, assoc_actor_2 (text) |
| **Evidence** | fatalities (integer); notes, source (text) |

## Example records from ACLED Database

| event_date | country | admin1 | event_type | actor1 | fatalities |
|---|---|---|---|---|---|
| 2023-02-01 | Georgia | Imereti | Protests | Protesters (Georgia) | 0 |
| 2023-02-01 | Georgia | Tbilisi | Protests | Protesters (Georgia) | 0 |

# Answering a Question vs. Research

- **Previous: the user supplies a specific question**
  - How did monthly battles in Colombia change between the pre-policy and policy periods?
- **<u>But, many times users do not know what question to ask</u>**
  - "Analyze Petro's total peace policy in Colombia and its impact on violence versus armed groups."
- **The agent should autonomously choose what questions to ask and how to explore on behalf of the user**

Published as a conference paper at COLM 2026

# DataSTORM: Deep Research on Large-Scale Databases using Exploratory Data Analysis and Data Storytelling

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https://arxiv.org/abs/2604.06474

# Knowledge Curation v. Data-driven Research

## Source: Internet

Knowledge Curation

- OpenAI Deep Research
- Gemini Deep Research
- STORM / Co-STORM
- Perplexity
- DR Tulu
- WebThinker



Lecture 2

## Source: Database

Data-driven research

- InsightBench / TablePilot
- Text-to-SQL agents

Few systems for end-to-end investigation

Quiz: Why might data-driven research be difficult?

# Why is deep research over databases hard?

Web sources often state findings. Database findings must be computed.

**WEB**



→ Retrieve + synthesize written evidence

**DATABASE**



→ Query + analyze to produce evidence

## 1 Which query comes next?

Open-ended goals leave a large space of possible queries and hypotheses.

## 2 What is hidden in the data?

Tables can exceed context; useful patterns may be absent from result previews.

## 3 Do the findings fit together?

Filters and time windows can drift; findings may lack a coherent argument.

**Only 23.3%** **of claims attributed to the database**

OpenAI DR (CSV) · ACLED benchmark

# Running Example: Colombia

"Analyze Petro's total peace policy in Colombia and its impact on violence versus armed groups."

- **ACLED event data**
  - Date, location, event type, actors, and fatalities
- **Web sources**
  - Policy dates, definitions, and context

# Exploratory Data Analysis

- **P1: Iterative hypothesis generation and testing**
  - Ask → analyze → use the result to choose the next question.
- **P2: Inductive discovery and deductive reasoning**
  - Deductive: top-down reasoning
  - Inductive: bottom-up surfacing
    - Statistics reveal patterns the agent did not anticipate.
    - Those patterns suggest hypotheses to test with targeted queries.

John W. Tukey. Exploratory Data Analysis. 1977
Andy Kirk. Exploratory data analysis: Using visuals to see your data. 2016
Hadley Wickham and Garrett Grolemund. Exploratory Data Analysis. 2017

# Data Storytelling

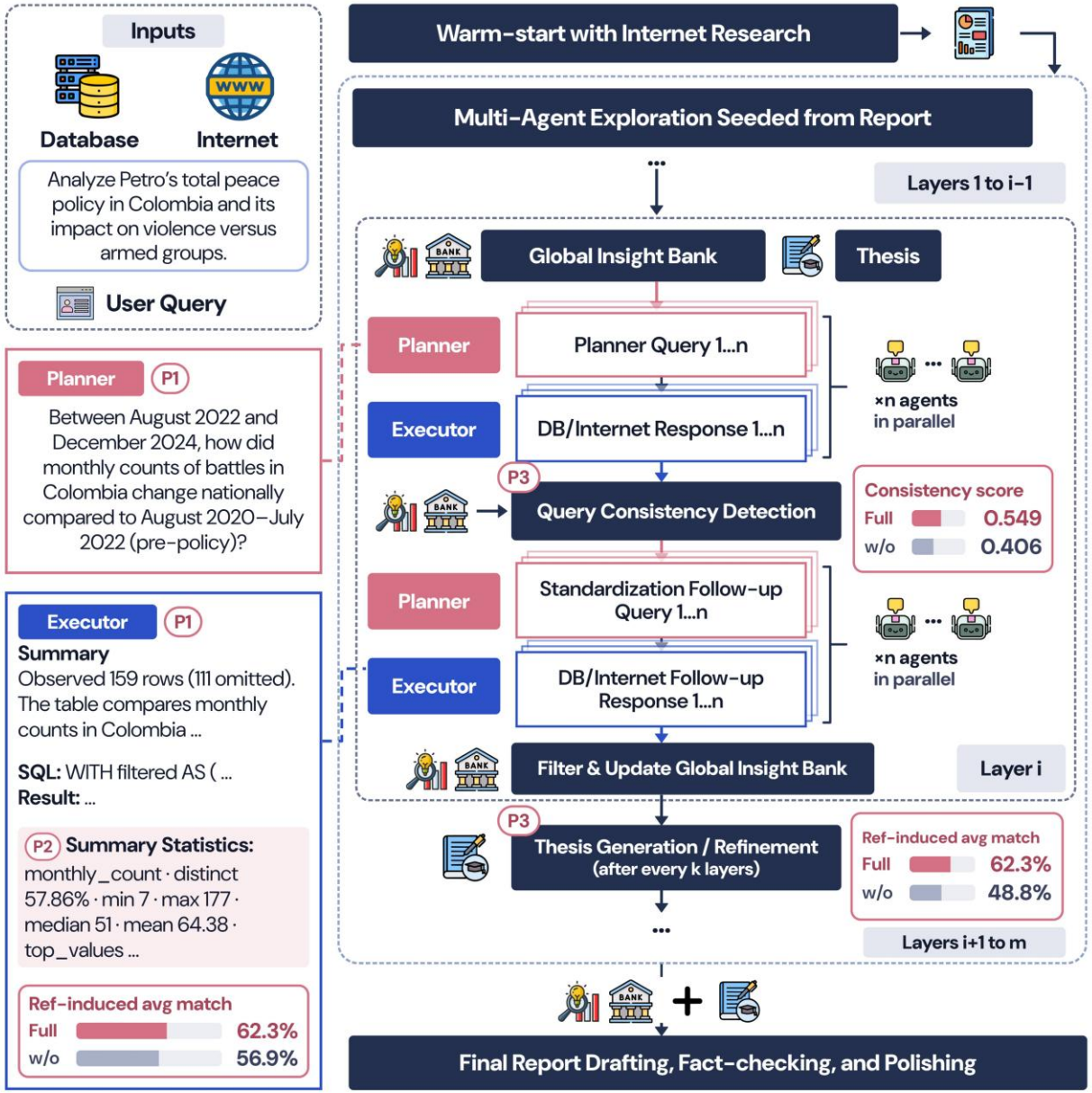
- **P3: Converge toward a central thesis**
  - Connect findings into a central argument.
  - Use it to guide exploration, then revise it with new evidence.

Edward Segel and Jeffrey Heer. Narrative visualization: Telling stories with data. 2010.
Brent Dykes. Effective Data Storytelling: How to Drive Change with Data, Narrative and Visuals. 2019.

# DataSTORM in Three Stages

- **1. Get context from the web**
  - Learn the necessary background and useful starting questions.
- **2. Investigate the database**
  - Ask questions, compute answers, and decide what to examine next.
- **3. Write the report**
  - Organize the findings around a thesis and check the supporting evidence.

# The DataSTORM Framework

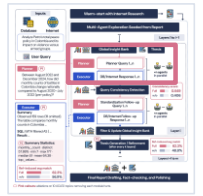



Pink callouts: ablations on 10 ACLED topics: removing each module hurts.

# What Happens in One Round?

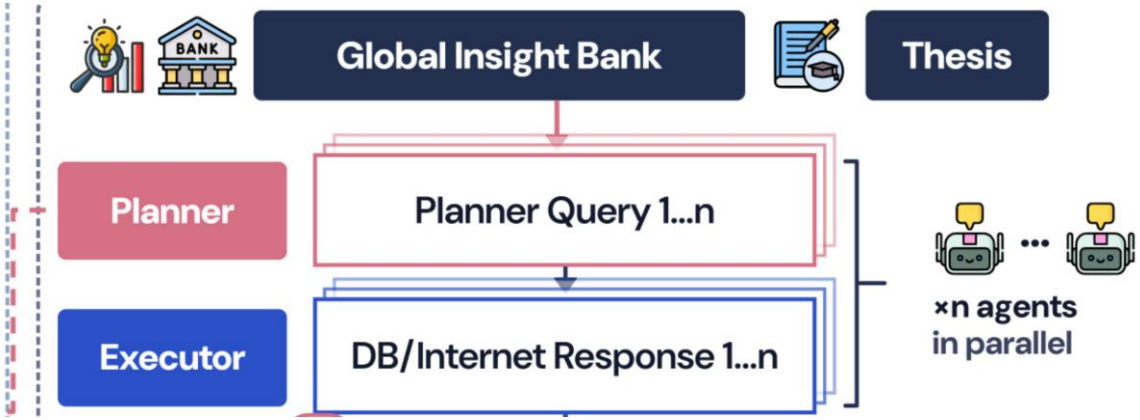
- **Planner: choose questions for the current round.**
- **Executors: return initial responses and the SQL used.**
- **Consistency module: compare the SQL definitions.**
  - Send corrective follow-ups to executors when needed.
    - Executors: return corrected queries and final responses.
- **Save useful insights in the insight bank.**
  - Use this shared evidence to plan the next round.

# Planner–executor decomposition



Framework

P1 Iterative hypothesis generation and testing




**Separate research planning from query execution.**

## Planner

Task: generate queries

## Executors

Decide how to answer the query

Explore data or web sources;
run queries and return answers.

Quiz: How should we implement the Executor?

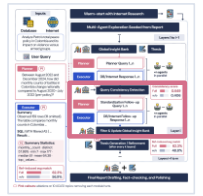
# Executor: SQL and a Result

The executor returns an answer grounded in an executed query.

| Response component | Response |
|---|---|
| Result size | 159 rows |
| Preview | 111 rows omitted |
| SQL | WITH filtered AS ( ... |
| Summary | The table compares monthly counts in Colombia ... |

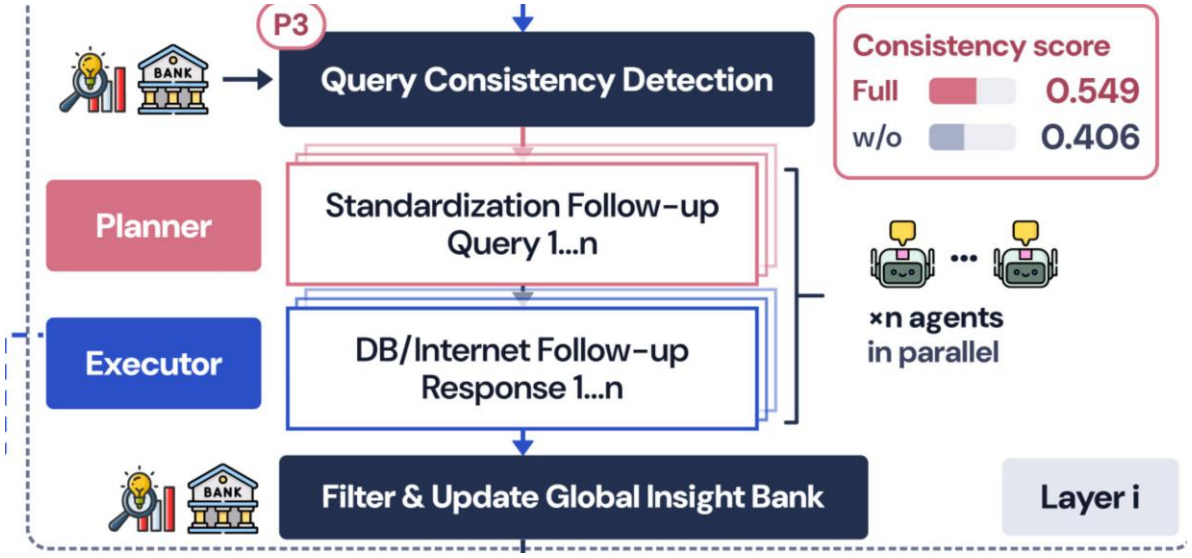
Datatalk/SPINACH

# Query consistency: align analytical definitions



Framework

P3 Convergence toward one coherent central narrative




## Compare SQL from executor responses

**Executor 1**

```
country = 'Colombia' AND
event_type = 'Battles'
```

**Executor 2**

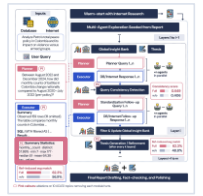
```
country = 'Colombia' AND
event_type = 'Battles' AND
sub_event_type = 'Armed clash'
```

Quiz: Would this lead to any problems?

# Standardization Follow-up

- **Start with the initial executor responses and their SQL.**
- **The consistency module identifies incompatible definitions.**
  - For this comparison: country, event type, periods and aggregation.
- **Send standardization follow-ups to the relevant executors.**
  - Use the original question and SQL as context.
- **Executors revise their queries and return follow-up responses.**
  - The resulting evidence can now be combined.

# Summary statistics: surface unexpected patterns



Framework

P2 Interplay of inductive discovery and deductive reasoning

**SQL result: 159 rows**

| | | |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
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| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |

## Compute over all 159 rows

**P2 Summary Statistics:**

monthly_count · distinct 57.86% · min 7 · max 177 · median 51 · mean 64.38 · top_values ...

# Statistics Reveal a Research Lead

Statistics for the returned monthly_count column

| Distinct values | Minimum | Maximum | Median | Mean |
|---|---|---|---|---|
| 57.86% | 7 | 177 | 51 | 64.38 |

- **The maximum, 177, is about 3.5 times the median, 51.**
- **High-count months may be missing from the preview.**
  - The statistics expose a lead that the planner can investigate.

**Deductive follow-up**

Next question

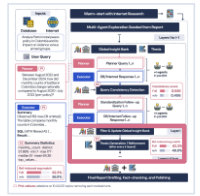
Which months account for the unusually high counts?

# Global Insight Bank

| Evidence | From the monthly-battle comparison |
|---|---|
| Question | National counts before and during the policy period |
| SQL and scope | Executed query; country, event type and date windows |
| Returned result | A preview of returned result |
| Statistical lead | Maximum 177; median 51; mean 64.38 |

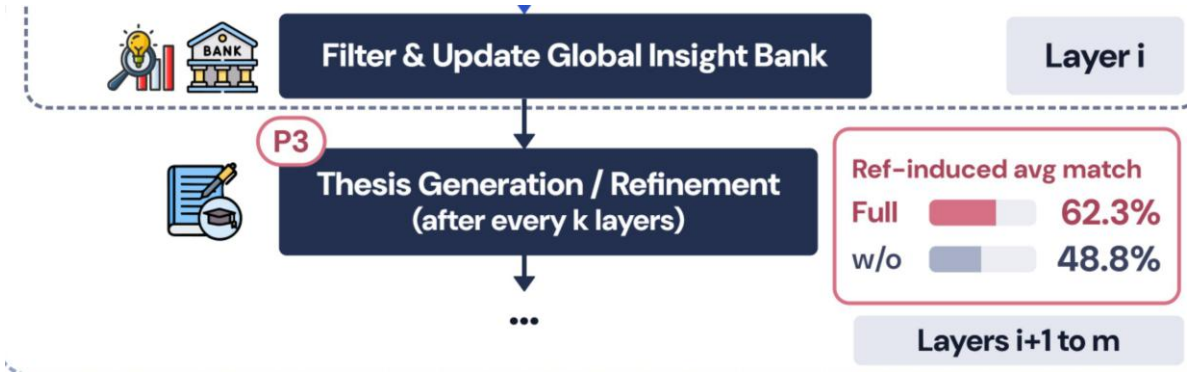
**At each layer, consolidate global insight bank by retaining up to a fixed number of insights.**

# Thesis generation and refinement



Framework

**P3** Convergence toward one coherent central narrative




**Sharpen**

Add specificity

**Pivot**

Change direction

**Confirm**

Retain the thesis

## Generate

Form a candidate thesis after initial exploration.

## Refine

Revisit it every k layers as new evidence accumulates.

# From Findings to a Thesis

- **Return to the research topic: the total peace policy.**
- **The monthly-battle comparison provides one line of evidence.**
  - Does violence differ consistently between the two periods?
  - Or is the pattern shaped by a few high-count months?
- **Build a central argument from the accumulated evidence.**
  - Refine it as further database and web findings arrive.

# Thesis event example

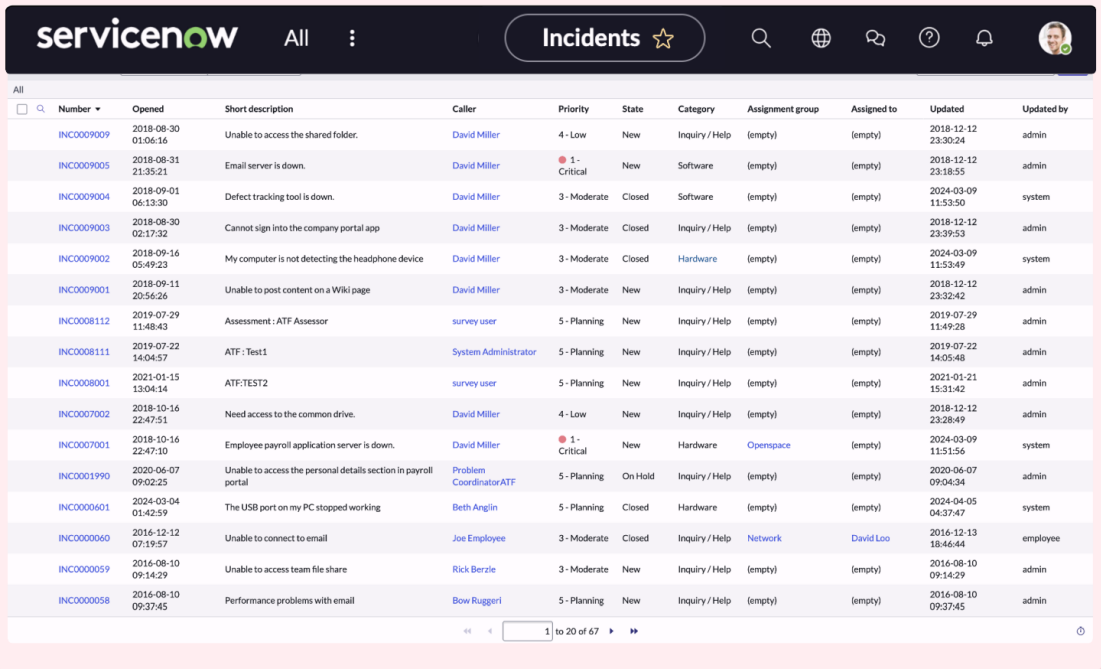
| Thesis Event | Thesis and added detail |
|---|---|
| Generate | Ceasefires without teeth,<br>violence with new faces |
| Sharpen | + New geographies |
| Sharpen | + Specified weapons and timing |

# Outline

- DataSTORM: Data-driven deep research
  - Motivation and running example
  - The agentic approach
  - **Evaluation**
- SEER-Repro: Automatic robustness checking
  - Reproduction and robustness
  - Results and expert review

# InsightBench: Business Analytics Benchmark

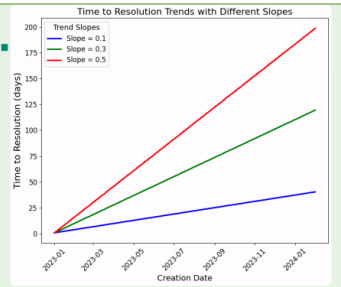
**a.**




| Number | Opened | Short description | Caller | Priority | State | Category | Assignment group | Assigned to | Updated | Updated by |
|---|---|---|---|---|---|---|---|---|---|---|
| INC0009009 | 2018-08-30 01:06:16 | Unable to access the shared folder. | David Miller | 4 - Low | New | Inquiry / Help | (empty) | (empty) | 2018-12-12 23:30:24 | admin |
| INC0009005 | 2018-08-31 21:35:21 | Email server is down. | David Miller | 1 - Critical | New | Software | (empty) | (empty) | 2018-12-12 23:18:55 | admin |
| INC0009004 | 2018-09-01 06:13:30 | Defect tracking tool is down. | David Miller | 3 - Moderate | Closed | Software | (empty) | (empty) | 2024-03-09 11:53:50 | system |
| INC0009003 | 2018-08-30 02:17:32 | Cannot sign into the company portal app | David Miller | 3 - Moderate | Closed | Inquiry / Help | (empty) | (empty) | 2018-12-12 23:39:53 | admin |
| INC0009002 | 2018-09-16 05:49:23 | My computer is not detecting the headphone device | David Miller | 3 - Moderate | Closed | Hardware | (empty) | (empty) | 2024-03-09 11:53:49 | system |
| INC0009001 | 2018-09-11 20:56:26 | Unable to post content on a Wiki page | David Miller | 3 - Moderate | New | Inquiry / Help | (empty) | (empty) | 2018-12-12 23:32:42 | admin |
| INC0008112 | 2019-07-29 11:48:43 | Assessment : ATF Assessor | survey user | 5 - Planning | New | Inquiry / Help | (empty) | (empty) | 2019-07-29 11:49:28 | admin |
| INC0008111 | 2019-07-22 14:04:57 | ATF : Test1 | System Administrator | 5 - Planning | New | Inquiry / Help | (empty) | (empty) | 2019-07-22 14:05:48 | admin |
| INC0008001 | 2021-01-15 13:04:14 | ATF:TEST2 | survey user | 5 - Planning | New | Inquiry / Help | (empty) | (empty) | 2021-01-21 15:31:42 | admin |
| INC0007002 | 2018-10-16 22:47:51 | Need access to the common drive. | David Miller | 4 - Low | New | Inquiry / Help | (empty) | (empty) | 2018-12-12 23:28:49 | admin |
| INC0007001 | 2018-10-16 22:47:10 | Employee payroll application server is down. | David Miller | 1 - Critical | New | Hardware | Openspace | (empty) | 2024-03-09 11:51:56 | system |
| INC0001990 | 2020-06-07 09:02:25 | Unable to access the personal details section in payroll portal | Problem CoordinatorATF | 5 - Planning | On Hold | Inquiry / Help | (empty) | (empty) | 2020-06-07 09:04:34 | admin |
| INC0000601 | 2024-03-04 01:42:59 | The USB port on my PC stopped working | Beth Anglin | 5 - Planning | Closed | Hardware | (empty) | (empty) | 2024-04-05 04:37:47 | system |
| INC0000060 | 2016-12-12 07:19:57 | Unable to connect to email | Joe Employee | 3 - Moderate | Closed | Inquiry / Help | Network | David Loo | 2016-12-13 18:46:44 | employee |
| INC0000059 | 2016-08-10 09:14:29 | Unable to access team file share | Rick Berzle | 3 - Moderate | New | Inquiry / Help | (empty) | (empty) | 2016-08-10 09:14:29 | admin |
| INC0000058 | 2016-08-10 09:37:45 | Performance problems with email | Bow Ruggeri | 5 - Planning | New | Inquiry / Help | (empty) | (empty) | 2016-08-10 09:37:45 | admin |

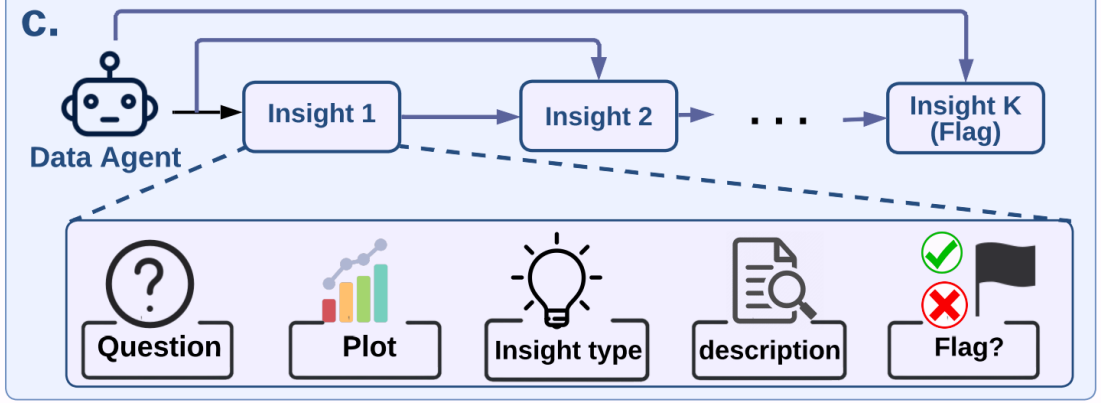
1 to 20 of 67

**b.**




```python
import pandas as pd
num_samples = 500
data_start_date = pd.to_datetime("2023-01-01")
data_end_date = pd.to_datetime("2024-02-01")
opened = np.random.choice(
    pd.date_range(start=data_start_date, end=data_end_date,
                  freq="min")[:-10],size=num_samples,
)
opened.sort()
ttr = [1 + slope * (created - data_start_date).days for created in opened]
```

**c.**



# InsightBench: Insight Recall

| Suppose these are the references | Best match |
|---|---|
| Maximum 177 is more than 3 × median 51. | 1.0 |
| One single month accounted for more than half of all casualties | 0.0 |

- **If the report covers only the first: (1 + 0) / 2 = 50%**

# Results on InsightBench

100 tasks: a goal, a database, and reference insights

| Recall (%) | Judge | AgentPoirot | DataSTORM |
|---|---|---|---|
| Insight | GPT-4o | 47.1 | **61.9** |
| Insight | Qwen3 | 49.9 | **69.3** |
| Summary | GPT-4o | 46.6 | **52.5** |
| Summary | Qwen3 | 51.5 | **58.7** |

Both systems use GPT-5. Judges score matches to references.

# Evaluation

| Question | Metric |
| --- | --- |
| Does the report cover relevant findings? | Reference-criteria match |
| How good is the report overall compared to expert reports? | RACE rubric; human preference |
| Does it use database evidence? | Database claim attribution |
| Is the research angle useful? | Journalist originality ratings |

# ACLED: Criteria Match (**Rubrics**)

An LLM turns the reference expert article into a checklist, then grades the report on each item.

| Criterion from the Colombia topic | Suppose the judge gives |
|---|---|
| Explain changes in state–group clashes | 1.0 — covered well |
| Competition between armed groups | 0.5 — partly covered |
| Geographic expansion of armed groups | 0.0 — omitted |

- **Average match: (1 + 0.5 + 0) / 3 = 50%.**
- Rubrics is a common form of eval metric today!

# RACE: expert article weighted score

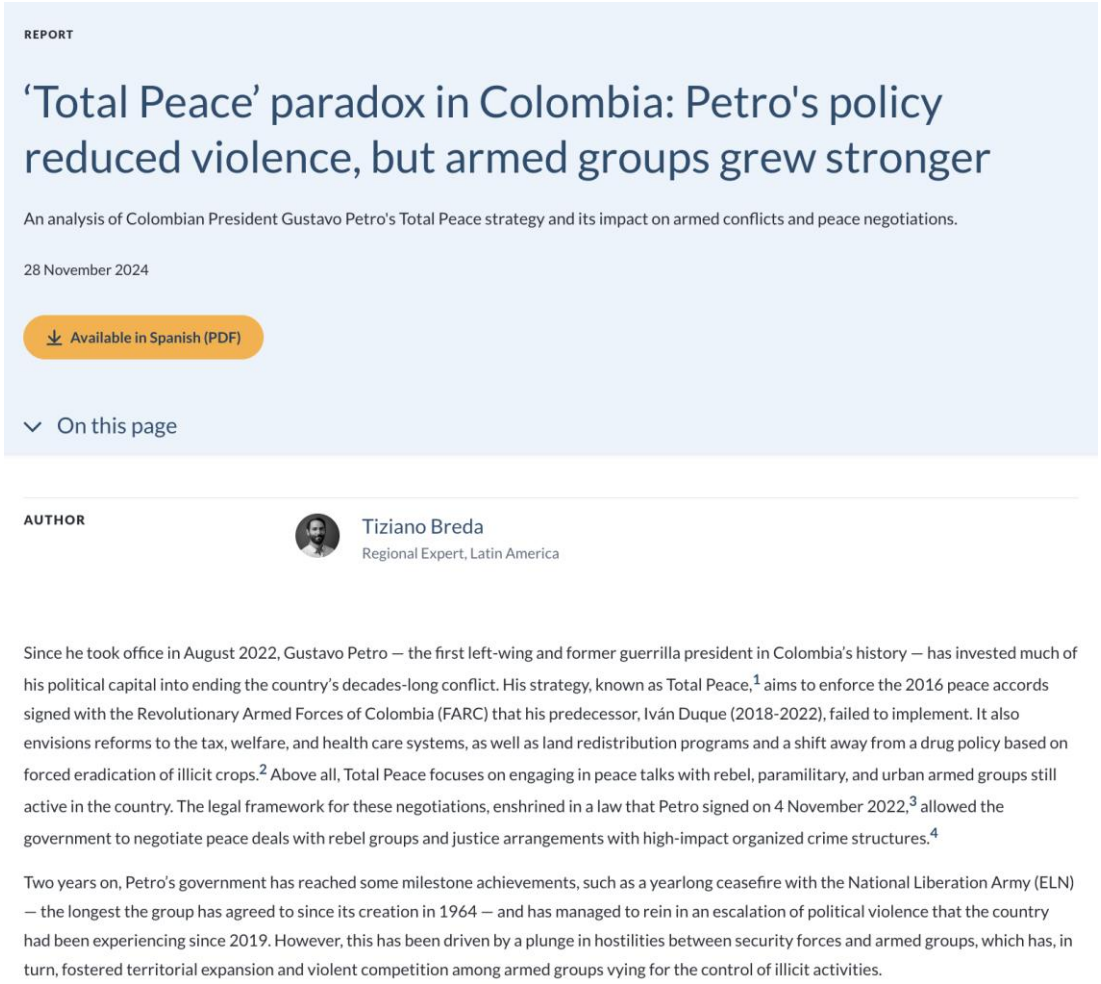
| Dimension | What it asks in the Colombia report |
|---|---|
| Comprehensiveness | Does it cover both violence and armed-group expansion? |
| Depth | Does it explain how the two patterns fit together? |
| Instruction following | Does it stay within the requested topic and dates? |
| Readability | Can the reader follow the evidence and argument? |

Relative score = 100 × report score / (report score + reference score)

If both weighted scores are 8: 100 × 8 / 16 = 50 (parity).

Du, M., Xu, B., Zhu, C., Wang, X., & Mao, Z. (2025). *DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents*. arXiv:2506.11763.

# Building the ACLED Benchmark



REPORT

## ‘Total Peace’ paradox in Colombia: Petro's policy reduced violence, but armed groups grew stronger

An analysis of Colombian President Gustavo Petro's Total Peace strategy and its impact on armed conflicts and peace negotiations.

28 November 2024

Available in Spanish (PDF)

On this page

AUTHOR

Tiziano Breda
Regional Expert, Latin America

Since he took office in August 2022, Gustavo Petro — the first left-wing and former guerrilla president in Colombia's history — has invested much of his political capital into ending the country's decades-long conflict. His strategy, known as Total Peace,[1] aims to enforce the 2016 peace accords signed with the Revolutionary Armed Forces of Colombia (FARC) that his predecessor, Iván Duque (2018-2022), failed to implement. It also envisions reforms to the tax, welfare, and health care systems, as well as land redistribution programs and a shift away from a drug policy based on forced eradication of illicit crops.[2] Above all, Total Peace focuses on engaging in peace talks with rebel, paramilitary, and urban armed groups still active in the country. The legal framework for these negotiations, enshrined in a law that Petro signed on 4 November 2022,[3] allowed the government to negotiate peace deals with rebel groups and justice arrangements with high-impact organized crime structures.[4]

Two years on, Petro's government has reached some milestone achievements, such as a yearlong ceasefire with the National Liberation Army (ELN) — the longest the group has agreed to since its creation in 1964 — and has managed to rein in an escalation of political violence that the country had been experiencing since 2019. However, this has been driven by a plunge in hostilities between security forces and armed groups, which has, in turn, fostered territorial expansion and violent competition among armed groups vying for the control of illicit activities.

Source: https://acleddata.com/report/total-peace-paradox-colombia-petros-policy-reduced-violence-armed-groups-grew-stronger

# Building the ACLED Benchmark

- **Start with 20 expert-written analyses**
  - Convert each into a neutral research topic
- **Give systems database and web access**
  - Keep the reference articles out of search results
  - Restrict web evidence to the reference publication date
- **Evaluate the resulting report against the reference**

Quiz: Why could using the article title reveal too much?

# Results on ACLED

20 topics · GPT-5 judge

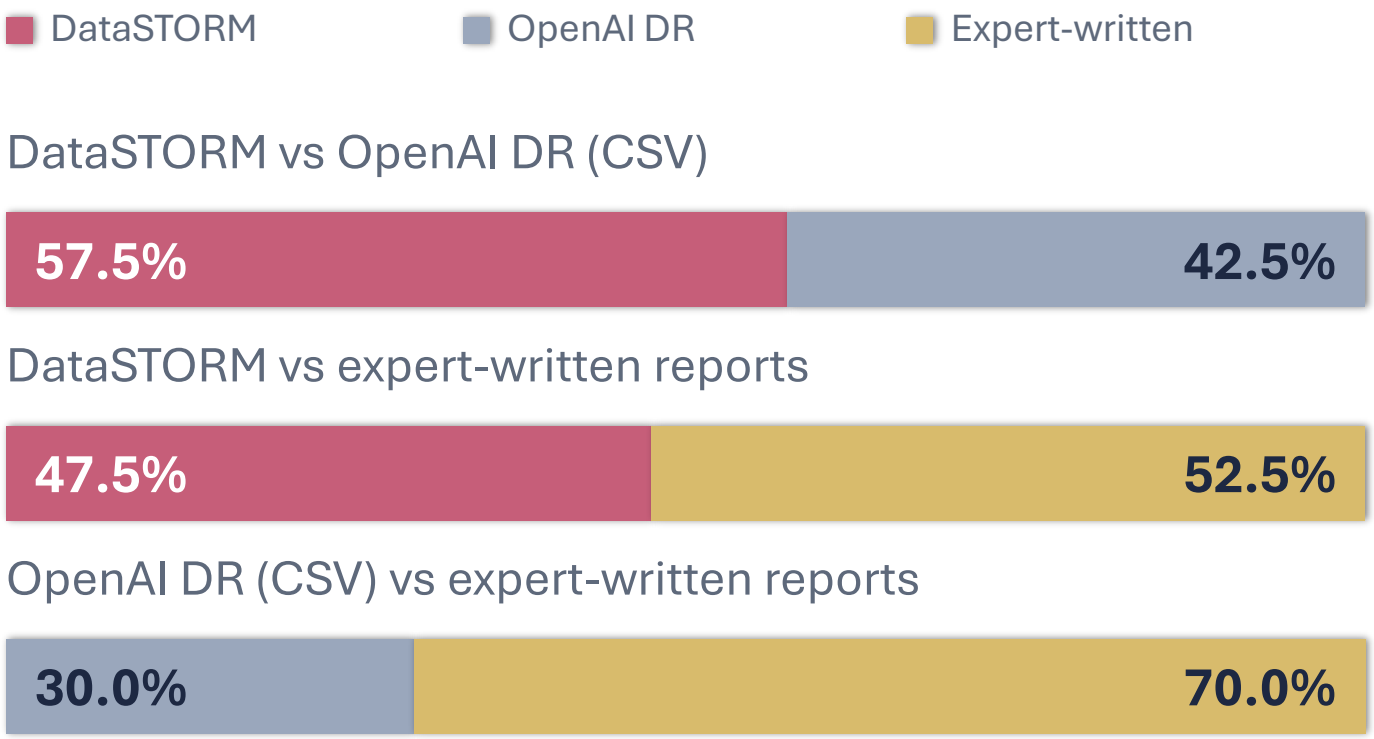
| System | Criteria match (%) | RACE | DB use (%) |
|---|---|---|---|
| OpenAI DR (MCP) | 48.5 | 46.1 | 30.4 |
| OpenAI DR (CSV) | 51.2 | 46.8 | 23.3 |
| **DataSTORM** | **61.8** | **52.6** | **66.4** |

MCP: database tools; CSV: file access

# Human Evaluation

10 journalists · 20 topics × 2 raters · system identities hidden during annotation

## Pairwise preference (% of comparisons)




## Originality

Mean rating · 1–5 scale

**2.21 → 3.05**

OpenAI DR DataSTORM

$p < 0.05$ · $N = 39$

# Ablations: What Do the Components Contribute?

| Removed component | Metric | Without | Full |
|---|---|---|---|
| Summary statistics | Criteria match | 56.9% | 62.3% |
| Thesis generation | Criteria match | 48.8% | 62.3% |
| Consistency checks | Consistency score | 0.406 | 0.549 |

10 ACLED topics · GPT-5 judge

# Outline

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  - Evaluation
- **SEER-Repro: Automatic robustness checking**
  - Reproduction and robustness
  - Results and expert review

# Trustworthy discovery needs two checks

**Before LLMs can be trusted to produce new scientific results,**
**They must first be able to reproduce existing studies**

## Reproducibility

**Can we recover the published result?**

Reconstruct the original analysis from the paper, data, and manuals.

## Analytical robustness

**Does the conclusion survive reasonable alternatives?**

Vary cohort definitions, endpoints, or statistical choices.

# AI-Driven Reproducibility and Analytical Robustness in Scientific Discovery

**Shicheng Liu[1]†, Krista Opsahl-Ong[2], Andrew Drozdov[2], Julian Killingback[4,2]**
**Arnav Singhvi[2], Ronak Pradeep[2], Michael Bendersky[2]**
**James C. Dickerson[3]*, Monica S. Lam[1]***

[1]Department of Computer Science, Stanford University
[2]Databricks AI Research
[3]Division of Oncology, Stanford University School of Medicine
[4]University of Massachusetts Amherst

https://arxiv.org/abs/2610.23845

# SEER: The backbone of U.S. cancer research

The NCI's Surveillance, Epidemiology, and End Results program has tracked cancer incidence and outcomes since 1973.

**1973 → today**

50+ years of population-based cancer surveillance

**126K+**

publications reference SEER

**≈25K studies**

used SEER data in primary analysis

## What SEER captures

| | |
|---|---|
| **Demographics** | Age, sex, race/ethnicity, geography |
| **Tumor** | Site, morphology, stage at diagnosis |
| **Outcomes** | Survival and vital status |

**39 tables · 800 pages of manuals**

## Why SEER fits this project

**High-impact literature**

Thousands of cancer-research questions and documented analytical choices.

**Computationally intensive**

Large databases and complex scientific analysis workflows.

**OUR USE** **A testbed for reproducing studies and assessing robustness.**

Source: NCI SEER Program overview and Annual Usage Metrics (2024).

# Findings and Atomic Claims

**Illustrative finding:**

"For cohort C over years Y, the crude rate rose from 4.0 to 5.0 per 100,000—a 25% increase."

- Starting rate: **4.0 per 100,000**
- Ending rate: **5.0 per 100,000**
- Relative increase: **25%**

# SEER-Repro makes published studies executable

A medical epidemiology testbed built from SEER cancer-registry studies.

**20**

published papers

**901**

empirical findings

**2,190**

atomic claims

## Execution environment

39 tables · 198.3M total rows
40 documentation resources

## Curated with domain experts

Historical data + population tables
Registry mappings + coding manuals

# Reproduction Approach

- **Extract empirical findings** from the paper
- **Reproduce each finding** with a coding agent
  - Use the paper, data, and manuals
  - Save the analysis code and execution results
- **Split findings into atomic claims and grade them**
  - Check method compatibility and evidence provenance
  - Reproduced / Close / Not reproduced

# Grading a Reproduction

First check evidence provenance and method compatibility.

$$\text{Relative error} = |\text{computed} - \text{reported}| \ / \ |\text{reported}|$$

| Reported | Computed | Error | Default verdict |
|---|---|---|---|
| 5.0 | 4.8 | 4% | Reproduced |
| 5.0 | 4.6 | 8% | Close |
| 5.0 | 4.4 | 12% | Not reproduced |

<= 5% Reproduced

5%-10% Close-to-reproduced

> 10% Not reproduced

# 77.4% of atomic claims reproduced

| Paper (shorter name) | Journal (year) | Cit. | Findings | Atomic claims | Reproduced | Close | Not reproduced |
|---|---|---|---|---|---|---|---|
| Bone sarcoma survival by tumor site (Hu et al., 2021) | Cancers (2021) | 37 | 56 | 188 | 187 (99.5%) | | |
| Sex disparities in cancer incidence (Cook et al., 2009) | Cancer Epidemiol. Biomarkers Prev. (2009) | 466 | 36 | 151 | 140 (92.7%) | | |
| Inflammatory breast cancer incidence and survival (Hance et al., 2005) | J. Natl. Cancer Inst. (2005) | 702 | 40 | 131 | 119 (90.8%) | | |
| Thyroid cancer trends (Davies & Welch, 2014) | JAMA Otolaryngol. Head Neck Surg. (2014) | 1,628 | 33 | 91 | 80 (87.9%) | | |
| Age at oropharyngeal cancer diagnosis (Cline et al., 2020) | Cancers (2020) | 29 | 75 | 154 | 129 (83.8%) | | |
| Adenoid cystic carcinoma of the breast (Ghabach et al., 2010) | Breast Cancer Res. (2010) | 188 | 51 | 128 | 103 (80.5%) | | |
| Adult diffuse glioma incidence (Li et al., 2018) | Cancer Med. (2018) | 72 | 60 | 111 | 89 (80.2%) | | |
| Childhood cancer incidence and mortality (Linet et al., 1999) | J. Natl. Cancer Inst. (1999) | 438 | 52 | 144 | 113 (78.5%) | | |
| Head and neck mucosal melanoma incidence (Marcus et al., 2012) | J. Skin Cancer (2012) | 83 | 20 | 54 | 42 (77.8%) | | |
| Esophageal adenocarcinoma incidence and mortality (Hur et al., 2013) | Cancer (2013) | 510 | 33 | 52 | 40 (76.9%) | | |
| Second cancers after young-onset colorectal cancer (He et al., 2018) | Cancer Med. (2018) | 43 | 59 | 124 | 95 (76.6%) | | |
| Lung cancer incidence by sex, race, and histology (Meza et al., 2015) | PLOS ONE (2015) | 368 | 36 | 111 | 80 (72.1%) | | |
| Age disparities in colorectal cancer incidence (Bailey et al., 2015) | JAMA Surg. (2015) | 1,102 | 27 | 82 | 59 (72.0%) | | |
| Fatal prostate cancer incidence by race (Kelly et al., 2017) | Eur. Urol. (2017) | 111 | 32 | 63 | 45 (71.4%) | | |
| Intracranial glioma incidence and mortality (Lin et al., 2021) | Front. Oncol. (2021) | 116 | 67 | 94 | 66 (70.2%) | | |
| Long-term mortality in young cancer survivors (Moke et al., 2021) | Cancers (2021) | 23 | 62 | 178 | 123 (69.1%) | | |
| Breast cancer incidence by age and tumor features (Jemal et al., 2007) | Breast Cancer Res. (2007) | 352 | 31 | 67 | 39 (58.2%) | | |
| Esophageal versus gastric cardia adenocarcinoma (El-Serag et al., 2002) | Gut (2002) | 266 | 34 | 51 | 29 (56.9%) | | |
| Metastatic cancer incidence and survival (Hudock et al., 2023) | Commun. Med. (2023) | 68 | 51 | 96 | 52 (54.2%) | | |
| Cervical cancer incidence by race and histology (Wang et al., 2004) | Cancer (2004) | 448 | 46 | 120 | 64 (53.3%) | | |
| **Total** | | | **901** | **2,190** | **1,694 (77.4%)** | **76 (3.5%)** | **420 (19.2%)** |

# Error Analysis

420 non-reproduced claims

| Cause | Claims | % |
|---|---|---|
| Data issues | 251 | 59.8 |
| Agent errors | 48 | 11.4 |
| Underspecified analysis | 47 | 11.2 |
| Unavailable software | 28 | 6.7 |
| Unresolved cause | 26 | 6.2 |
| Suspected evaluation errors | 20 | 4.8 |

Quiz: What does this suggest about the challenges around reproduction?

# A Reproduced Result

Suppose a study reports: “Cancer incidence doubled among younger adults.”

| Rate per 100,000 | Earlier | Later |
|---|---|---|
| Reported and reproduced | 20 | 40 |

- **The reproduction matches the reported numbers.**
- **But how was “younger adults” defined?**

# Two Definitions of "Younger Adults"

The upper age limit determines which records enter the analysis.

| Definition | Included ages |
|---|---|
| Definition 1 | 20–49: people in their 20s, 30s and 40s |
| Definition 2 | 20–39: people in their 20s and 30s |

The reported result used Definition 1.

# Quiz 1: Is This Important?

- **Would the age cutoff matter to how you interpret "incidence doubled"?**
- What might a reader infer about adults in their 20s and 30s?
- Would reproducing the original numbers settle that question?

# Quiz 2: How Do We Test It?

- **How would you test whether the conclusion depends on the age cutoff?**
- What would you change?
- What would you hold fixed?
- Which outputs would you compare?

# Change One Definition and Rerun

- **Validate the baseline: ages 20–49 gives 20 → 40.**
- **Change only the upper age cutoff: 49 → 39.**
  - Use the matching population denominator for ages 20–39.
- **Keep the rest of the analysis fixed.**
  - Cancer definition, periods, registries and rate calculation.
- **Compare the rates, relative change and interpretation.**

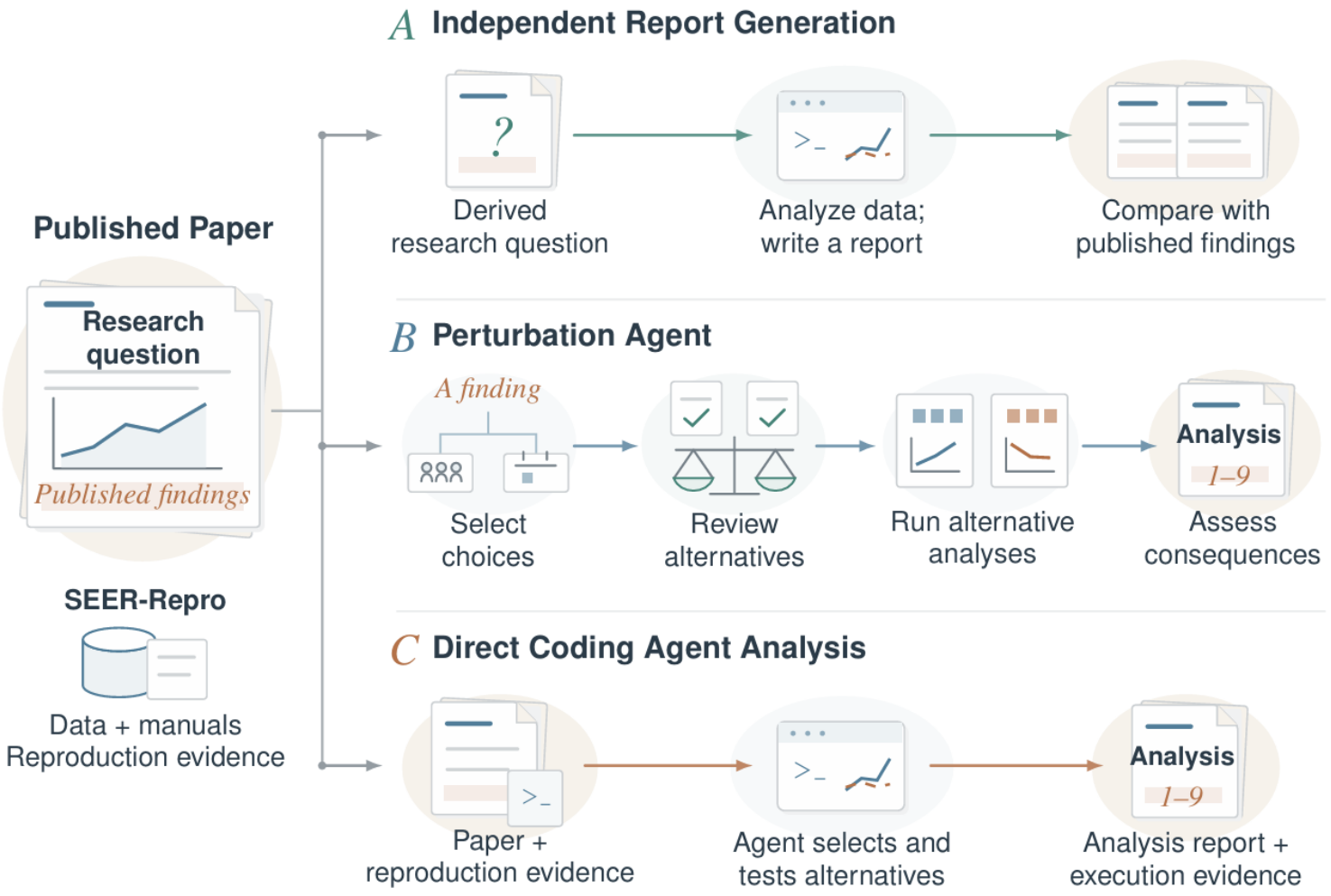
# The Increase Is in Ages 40–49

Each age band has 100,000 people in each period. Rates are per 100,000.

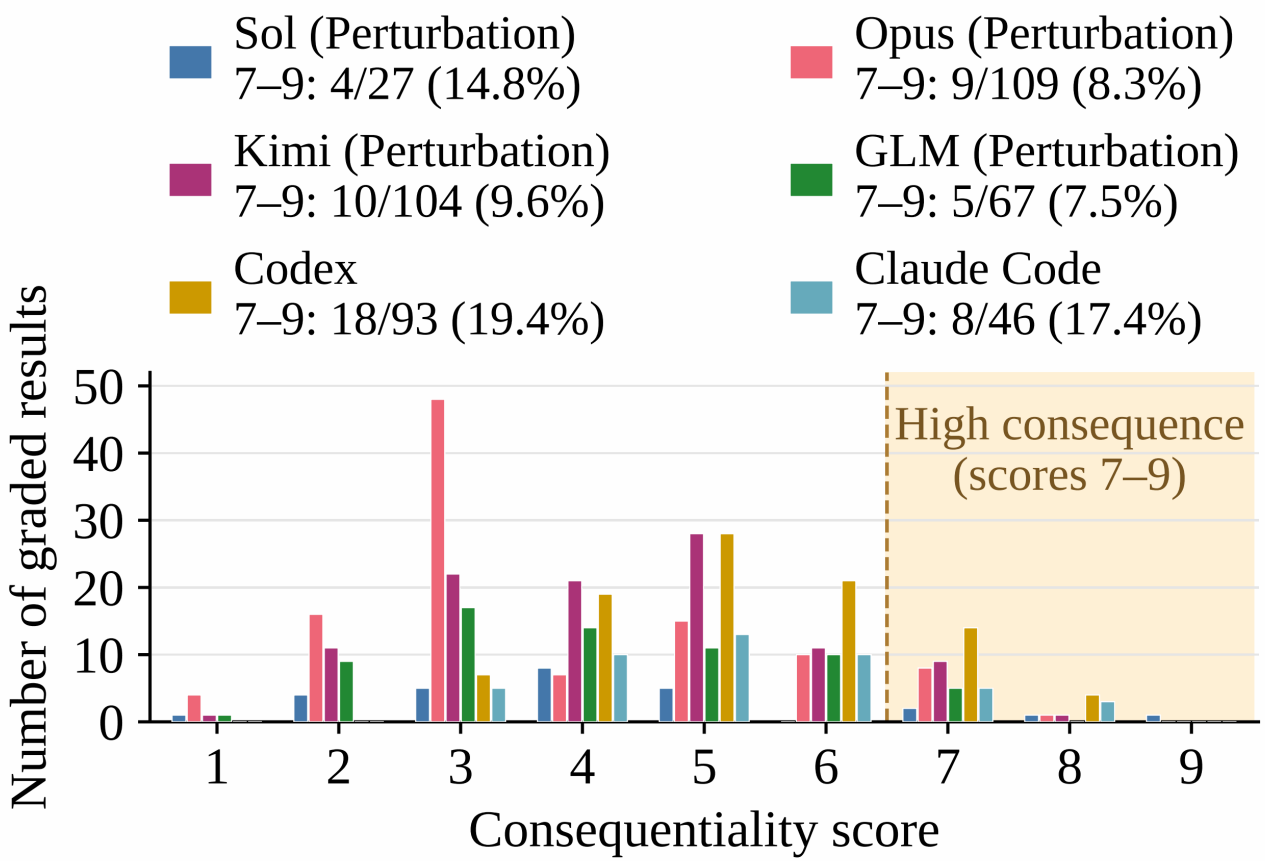
| Population | Earlier rate | Later rate |
|---|---|---|
| Ages 20–39 | 20 | 20 |
| Ages 40–49 | 20 | 60 |
| Definition 1: ages 20–49 | 20 | 40 |
| Definition 2: ages 20–39 | 20 | 20 |

Definition 1: +100%.  Definition 2: no increase.

# How SEER-Repro Tests Alternatives



# 55 of 579 findings flagged for expert review




**9.5%**

55 / 579 findings flagged by ≥1 analyst

**2.4%**

14 / 579 findings flagged by ≥2 analysts

**Do different analysts complement each other?**

**74.5%**

41 of 55 affected findings identified by only one analyst

Suggests largely complementary sets of potentially consequential analytical choices.

# Expert Review

One expert reviewed 10 selected high-score cases.

| Assessment | Cases |
|---|---|
| Alternatives likely less defensible | 2 |
| Limited qualifications | 7 |
| Further review needed | 1 |